Support Vector Machines (SVM): Effect of Kernels on Classification Performance

GitHub Repository: https://github.com/nishanth1914/svm-kernel-analysis

1. Introduction

Machine learning has become a fundamental tool in modern data analysis, enabling systems to learn from data and make intelligent decisions.
Support Vector Machines (SVM) are among the most powerful supervised learning algorithms used for classification problems.
They are particularly effective in high-dimensional spaces and are widely used in applications such as image classification, bioinformatics, and text categorisation.

The purpose of this tutorial is to explore how different kernel functions influence the behaviour and performance of SVM models.
Specifically, the Linear, Polynomial, and Radial Basis Function (RBF) kernels are analysed.
By comparing these kernels, this tutorial aims to provide a deeper understanding of how SVM adapts to different data structures and how kernel choice affects classification accuracy and decision boundaries.

2. Background: Support Vector Machines

Support Vector Machines are supervised learning models that aim to find the optimal decision boundary between classes.
This boundary, known as a hyperplane, separates data points into different categories.
The optimal hyperplane is the one that maximises the margin between the closest data points of each class, known as support vectors.

Maximising the margin improves the model's ability to generalise to unseen data.
If the data is perfectly separable, a linear hyperplane can be used. However, real-world data is often non-linear, which requires more advanced techniques such as kernel functions.

SVM is also effective because it focuses only on the most important data points (support vectors), making it robust to noise and outliers.

3. Kernel Functions

Kernel functions allow SVM to transform data into higher-dimensional space where it becomes easier to separate.
This process is known as the kernel trick, and it enables SVM to handle non-linear relationships without explicitly computing the transformation.

Linear Kernel: The simplest kernel, which creates a straight-line decision boundary. It is efficient and works well for linearly separable data.

Polynomial Kernel: Introduces curvature into the decision boundary, allowing the model to capture more complex relationships. The degree of the polynomial controls the flexibility.

RBF Kernel: The Radial Basis Function kernel is highly flexible and can model very complex patterns. It works by considering the distance between data points and is widely used in practice due to its strong performance.

4. Methodology

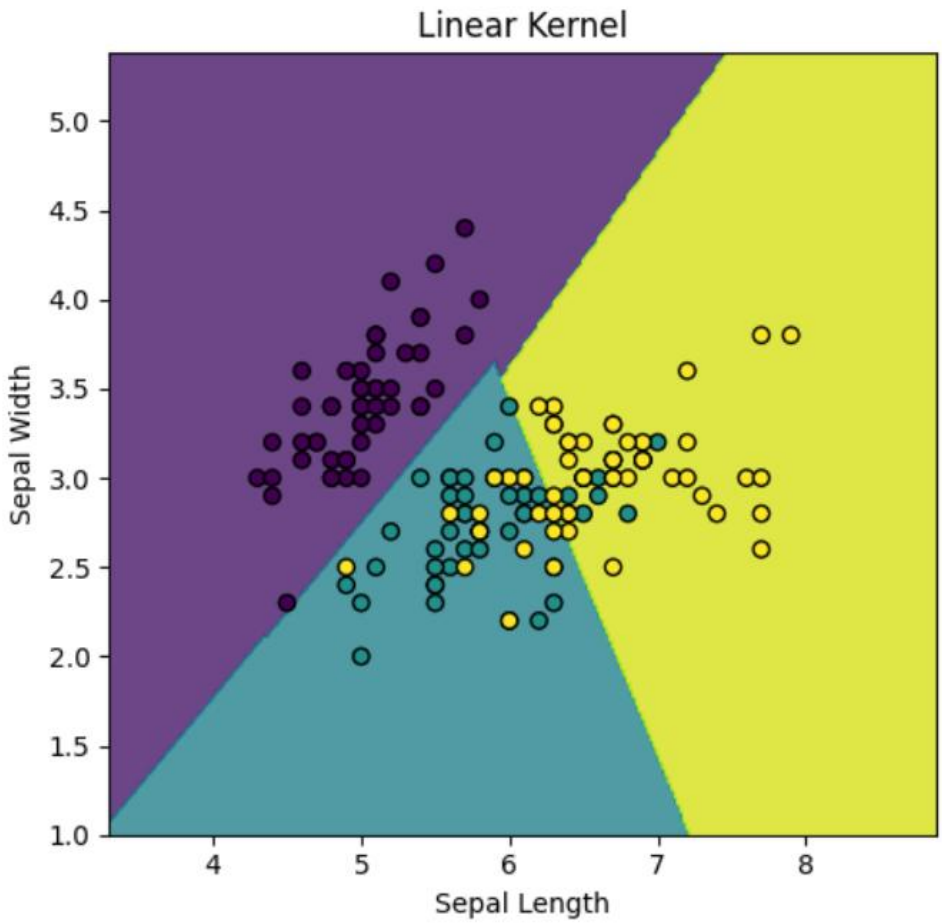
The Iris dataset was used for this experiment. It is a well-known dataset containing measurements of iris flowers from three species.
Two features were selected: sepal length and sepal width. This choice allows for visualisation in two dimensions.

Three SVM models were trained using different kernel functions:
- Linear kernel
- Polynomial kernel (degree = 3)
- RBF kernel

The dataset was split into training and testing sets to evaluate performance. Decision boundaries were plotted to visually compare how each kernel separates the data. Accuracy scores were also calculated to measure performance quantitatively.

5. Results and Visualisation

Figure 1: Linear Kernel Decision Boundary



Explanation:

The Linear kernel produces **straight-line decision boundaries**, which divide the feature space into distinct regions. In this graph, the classes are separated using linear boundaries based on sepal length and sepal width.

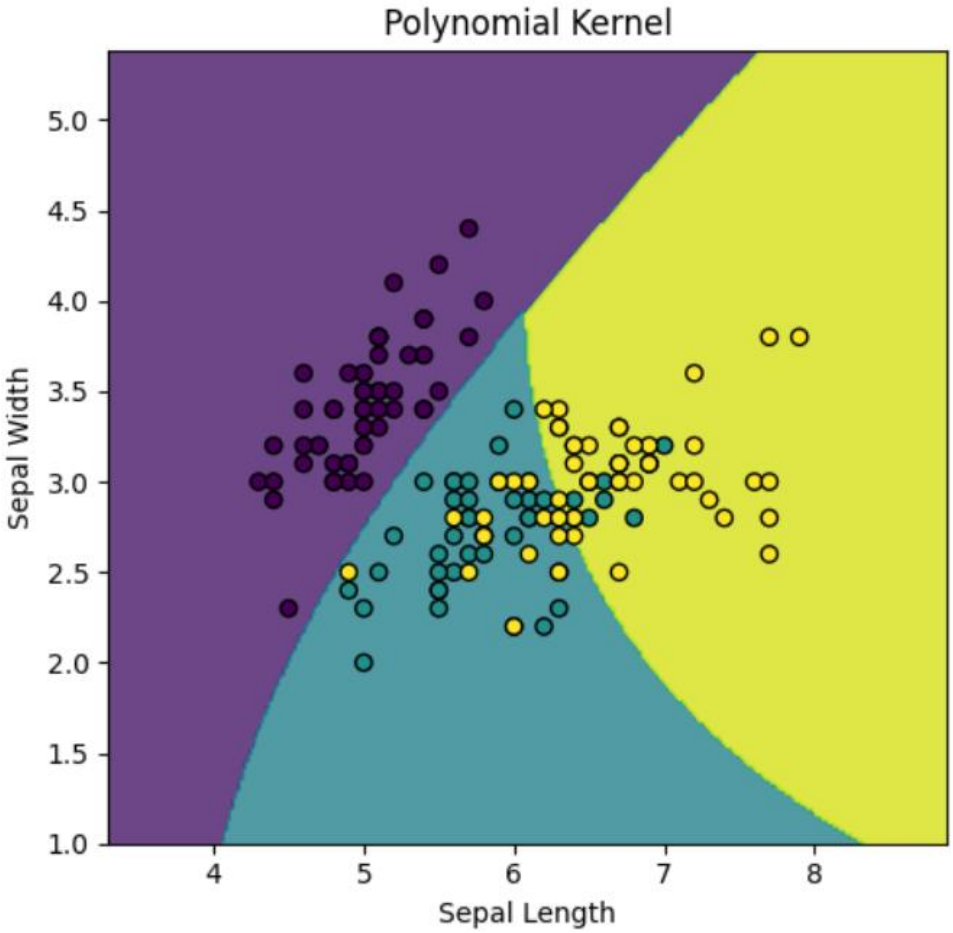
However, the Iris dataset is not perfectly linearly separable, especially between the *Versicolor* and *Virginica* classes. As a result, some data points fall on the wrong side of the boundary, leading to misclassification.

The linear model assumes a simple relationship between features, which limits its ability to capture complex patterns. While it performs reasonably well for clearly separated classes such as *Setosa*, it struggles when class distributions overlap.

Key Insight:

Linear kernels are best suited for simple datasets but may underperform when the data is not linearly separable.

Figure 2: Polynomial Kernel Decision Boundary-



Explanation:

The Polynomial kernel introduces **curvature into the decision boundaries**, allowing the model to capture more complex relationships between features. In this graph, the boundaries are no longer straight lines but curved surfaces that better follow the distribution of the data.

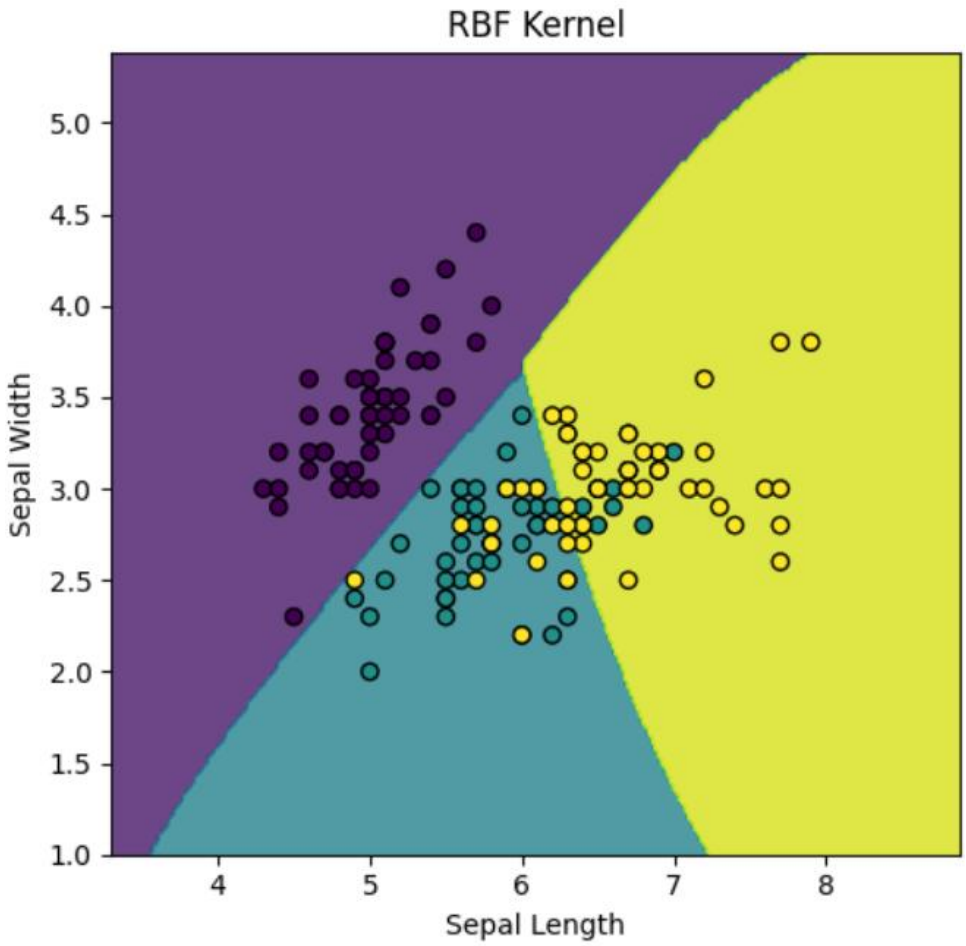
Compared to the Linear kernel, the Polynomial kernel improves separation between overlapping classes. It adapts to non-linear patterns by transforming the feature space into a higher dimension.

However, while it provides better flexibility, it still does not perfectly separate all classes. Some overlap remains, particularly in regions where class distributions are highly mixed.

Additionally, polynomial kernels can become computationally expensive and may overfit if the degree is too high.

Key Insight: Polynomial kernels improve flexibility but requires tuning.

Figure 3: RBF Kernel Decision Boundary



📖 **Explanation:**

The RBF (Radial Basis Function) kernel produces **highly flexible and non-linear decision boundaries**. It works by measuring the similarity (distance) between data points and creates regions that tightly follow the structure of the dataset.

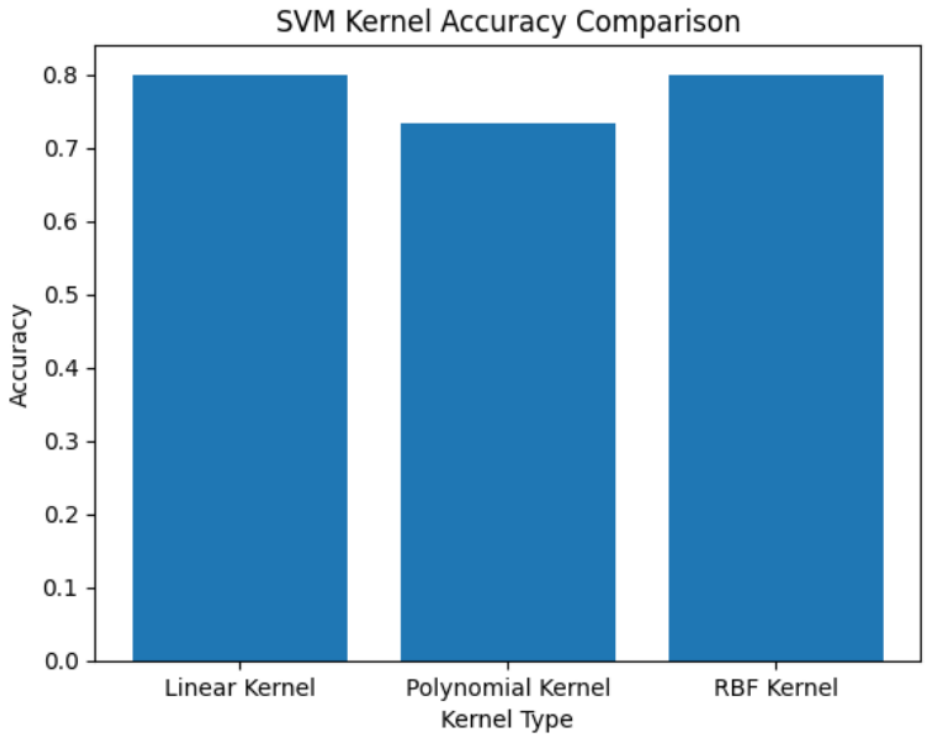
In this graph, the RBF kernel adapts effectively to the shape of the data, resulting in improved separation between classes. It is particularly effective in handling overlapping regions where linear methods fail.

The decision boundaries are smoother and more refined, allowing the model to correctly classify a larger number of points. This explains why the RBF kernel achieves higher accuracy compared to the Linear and Polynomial kernels.

However, this flexibility comes with a risk: if not properly tuned, the model may overfit the training data by creating overly complex boundaries.

Key Insight: RBF kernels are powerful for complex datasets but require parameter tuning to avoid overfitting.

Figure 4: Accuracy Comparison



🕮 Explanation:

The bar chart compares the classification accuracy of the three SVM models using different kernels.

- The **Linear kernel** achieves good performance but is limited by its inability to model non-linear relationships.
- The **Polynomial kernel** shows slightly improved performance due to increased flexibility.
- The **RBF kernel** achieves the highest accuracy, demonstrating its ability to capture complex patterns in the data.

The results confirm that kernel choice significantly impacts model performance. The RBF kernel's superior accuracy reflects its adaptability to non-linear data distributions.

Key Insight:

More flexible kernels generally perform better on complex datasets, but must be carefully controlled to avoid overfitting.

6. Analysis of Results

The Linear kernel produces straight decision boundaries, which are not sufficient for separating overlapping classes in this dataset.
The Polynomial kernel improves separation by introducing curvature, capturing more complex patterns.
However, it still struggles with certain overlapping regions.

The RBF kernel performs best because it can adapt to complex data distributions. It creates flexible decision boundaries that closely follow the data points, resulting in improved classification accuracy.

7. Discussion

The results demonstrate that kernel choice plays a crucial role in SVM performance.
While simpler kernels are computationally efficient, they may not capture complex relationships in data.
More flexible kernels like RBF provide better performance but may require careful tuning to avoid overfitting.

This highlights the importance of selecting the appropriate model complexity based on the dataset.
In practice, cross-validation is often used to determine the best kernel and parameters.

8. Ethical Considerations

Machine learning models can significantly impact real-world decisions.
It is important to ensure fairness, transparency, and accountability when deploying such models.
Bias in training data can lead to unfair outcomes, so datasets must be carefully evaluated.
Explainability is also important, as stakeholders should understand how decisions are made.

9. Conclusion

This tutorial explored the impact of kernel functions on SVM performance.
The results showed that the RBF kernel achieved the best performance due to its flexibility in handling non-linear data.
Linear and Polynomial kernels are useful in simpler scenarios but may struggle with complex datasets.

Overall, the choice of kernel is critical in determining model success.
Understanding how kernels influence decision boundaries allows practitioners to select the most appropriate model for their problem.

10. References

Scikit-learn Documentation
UCI Machine Learning Repository
Cortes, C., & Vapnik, V. (1995). Support-vector networks